Ans. (d) : The acceleration due to gravity of catastrophic earthquake will be > 980 cm/sec^2.

50. When it is noon along 82° 30' E longitude along what degree of longitude it will be at 6.30 am ?

(a) along 165° E
(b) along 67°30' W
(c) 0° E or W
(d) along 82° 30' W

Ans. (c) : When it is noon along 82^0 30' E longitude
i.e. at the time 12:00 pm = 82^0 30' E
Degree of longitude 6:30 am = ?
12:00 – 6:30 = 5:30 h
∵ 1 longitude is equal to 4 minute

$$\therefore 5{:}30 \text{ h} = \frac{5\times 60+30}{4} = \frac{330}{4} = 82.5 = 82^0.30'$$

From solving we prove that at the time of 6:30 am the longitude = 0^0 mid line

51. The first bird is believed to have appeared on the surface of the earth

(a) 300 million years ago
(b) 200 million years ago
(c) 150 million years ago
(d) 100 million years ago

Ans. (c) : Birds are one of the most recognizable and diverse groups of modern vertebrates. Over the past two decades, a wealth of new fossil discoveries and phylogenetic and macro evolutionary studies had transformed our understanding of how birds originated and became so successful. Birds evolved from theropoda dinosaurs during the Jurassic (around 165-150 million years ago) and their classic small, lightweight, feathered and winged body plan was pieced together gradually over tens of millions of years of evolution rather than in one burst of innovation.

52. An aeroplane takes off from 30° North Latitude, 50° East Longitude, and lands at the opposite end of the earth. Where does it land ?

(a) 30° North Lat., 50° West Long
(b) 30° South Lat., 50° West Long
(c) 50° North Lat., 30° West Long
(d) 30° South Lat., 130° West Long

Ans. (a) : When a aeroplane takes off from 30^0 North Latitude, 50^0 East Longitude, and lands at the opposite end of the earth then its landing position is 30^0 North Latitude & 50^0 West Longitude because to go from one hemisphere to the other the airplane has to pass through the poles.

53. Which one of the following is in the correct sequential order in which these appeared under simulated primitive earth condition ?

(a) Methane, Hydrogen cyanide, nitriles, aminoacids
(b) Hydrogen cyanide, methane, nitriles, aminoacids
(c) Aminoacids, nitriles. Hydrogen cyanide, methane
(d) Nitriles, aminoacids, methane, Hydrogen cyanide

Ans. (a) : In the primitive land conditions simulated in the earth, complex organic compounds gradually emerged from simple organic compounds. In this sequence, first methane gas, then hydrogen cyanide, nitrile and amino acids were produced. It is known that with the help of amino acids, the first organism originated in water.

54. The shortest air route from Perth to London is

(a) Perth, Bombay, Rome, London
(b) Perth, Ankara, Paris, London
(c) Perth, Aden, Paris, London
(d) Perth, Mombasa, Rome, London

Ans. (b) : The distance between London and Perth-8989 miles (or- 14466 km).
The shortest air route Perth-Ankara, Paris - London.

55. Given below is a map of America: The shaded (marked 1 and 2), dotted (marked 3) and the plain areas (marked 4) indicate respectively, the regions where the languages spoken are



(a) Spanish, French, Portuguese and others
(b) French, Spanish, Portuguese and others
(c) French, Portuguese, Spanish and English
(d) Portuguese, French, Spanish and others

Ans. (c) : From the map of America (South America & North America). The shaded area marked (1) is the Eastern coast of Canada where French is spoken.
The shaded area marked (2) is the area of Brazil, where Portuguese is spoken.
The shade marked (3) is the area of Argentina & Mexico where Spanish is spoken and the shaded marked (4) is the area of Canada & USA where English is spoken.

56. Consider the map given below :



The map shows the sea voyage route followed by

(a) Thomas Cook (b) Ferdinand Magellan
(c) John Cabot (d) Vasco da Gama

Ans. (b) : In search of fame and fortune, Portuguese explorer Ferdinand Magellan (C 1480-1521) set out from Spain in 1519 with a fleet of five ships to discover a western sea route to the Spice Islands. En route he discovered what is now known as the strait of Magellan and became the first European to cross the Pacific Ocean.
The voyage was long and dangerous, and only one ship returned home three years later.

INDIAN CONSTITUTION AND POLITY

57. The basic structure theory of the Constitution of India implies that

(a) certain features of the Constitution are so essential to it that they cannot be abrogated
(b) fundamental rights cannot be abridged or taken away
(c) the Constitution cannot be amended except in accordance with the procedure prescribed in Art, 368
(d) the Preamble of the Constitution cannot be amended for it is not a part of the Constitution and at the same time represents its real spirit

Ans. (a) : The basic structure theory of the constitution of India implies that the certain features of constitution are so essential to it that they cannot be abrogated.

58. Freedom of the Press in India

(a) is specifically provided in Article 19 (1)(a) of the Constitution
(b) is implied in the wider freedom of expression guaranteed by Article 19(1)(a) of the Constitution
(c) is guaranteed under the provisions of Article 361 A of the Constitution
(d) emanates from the operation of the Rule of Law In the country

Ans. (b) : Freedom of the Press in India is implied in the wider freedom of expression guaranteed by Article 19(1)(a) of the constitution. Article 19(1)(a) says that all citizens shall have the right to freedom of speech and expression.

59. Who among the following are appointed by the President of India ?

I. The Chairman, Finance Commission .
II. The Deputy Chairman, Planning Commission
III. The Chief Minister of a Union Territory.
Choose the correct answer from the codes given below.
Codes:

(a) I only (b) I and II only
(c) I and III only (d) II and III only

Ans. (c) : From the question the Chairman, Finance Commission and the Chief Minister of a Union Territory are appointed by the President of India. The Finance Commission is a constitutionally mandated body that is at the centre of fiscal federalism. It was set up under Article 280 of the constitution.
By the Central Government Act Section 45 in the Government of Union Territories Act, 1963 the Chief Minister shall be appointed by the president and the other ministers shall be appointed by the President on the advice of the Chief Ministers.

60. Assertion (A): In India, the political parties which formed the governments represented the majority of seats secured in the elections to the House of the People at the Centre and the Legislative Assemblies in the States but not the majority of votes.
Reason (R): The elections based on the majority vote system decide the result on the basis of relative majority of votes secured.
In the context of the above statements, which one of the following is correct?

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not a correct explanation of A